

SPECTRE SYSTEMD PERSISTENCE

Campaign-specific hunting for the source-observed hardware-monitor.service unit on Defender-onboarded Linux servers.

HUNTING QUERY

CONFIDENCE: MEDIUM

hardware-monitor.service

Before-sysinit target

acpi_pad.ko

source-reported disguise

Observable boundary

file path + device + writer context

REPORT DATE

27 August 2026

STATUS

Untested implementation sketch

PRIMARY TELEMETRY

Defender XDR /
DeviceFileEvents

ATT&CK

T1543.002 - Systemd
Service

A filename match indicates exposure requiring investigation. It does not prove SPECTRE execution, rootkit loading or compromise.



EXECUTIVE ASSESSMENT

Decision-grade summary and analytical boundaries.

ASSESSMENT

OPERATIONAL VALUE

9.1

Immediate exposure check for Linux servers potentially affected by UAT-10147.

CLASSIFICATION

Campaign-specific artifact

Hunting query - Untested implementation sketch. Not a durable behavioral rule.

1selected item
0production rules
Mediumconfidence
MDEprimary platform

Source fact

Talos reports persistence through hardware-monitor.service, description Hardware Performance Monitor, ordered before sysinit.target.

Analytical meaning

A matching unit file is a strong campaign pivot, but does not establish its content, enablement, module load or compromise.

Assumptions

Defender for Endpoint is present, Linux file events are retained, and the actor has not renamed the unit.

Promotion boundary

Do not alert until controlled writes prove coverage and legitimate ownership is understood.



THREAT MECHANICS

Source-supported chain and observable boundaries.

BEHAVIOR

01 Linux implant

SPECTRE executes on a compromised server.
→

02 Unit creation

hardware-monitor.service is placed in a system unit directory.
→

03 Early boot

Before=sysinit.target orders execution before normal initialization.

04 Rootkit disguise

Module appears as acpi_pad.ko.
→

05 Kernel control

Talos documents hiding and privilege functions.
→

06 Hunt boundary

This query observes only the unit file path.

Claim	Support	Used in query
Unit filename	Direct Talos observation	Yes
System unit directories	Deployment variants	Yes, declared assumption
Service content / boot order	Source fact, not file telemetry	No
Rootkit behavior	Source analysis	No

ATT&CK

T1543.002 - Create or Modify System Process: Systemd Service. No other technique is assigned to the hunt.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and platform limits.

TELEMETRY

REQUIRED

DeviceFileEvents

Microsoft Defender for Endpoint on Linux, with file-event visibility for system unit directories.

OPTIONAL ENRICHMENT

Initiating process, account, SHA1, file size, CMDB/package ownership, internet exposure and Live Response retrieval.

Field	Reliability	Purpose
Timestamp, DeviceId, DeviceName	Documented	Event and device pivot
FileName, FolderPath	Documented	Primary artifact filter
ReportId	Documented	Event uniqueness with device/time
ActionType	Documented; values unverified	Output only, never filtered
InitiatingProcess*	Documented; environment-dependent	Writer context
SHA1, FileSize	Documented; may be empty	Enrichment

Unavailable through this hunt

Unit content, enablement, Before=sysinit.target, kernel-module load, rootkit hooks and C2 success.

Sensor dependency

Microsoft notes that unsupported/disabled eBPF configurations can miss specific file and socket-related events. A clean result is not proof of absence.



DETECTION LOGIC

Simple artifact hunt with no unsupported threshold or correlation.

KQL

HYPOTHESIS

If an attacker deploys the reported persistence, file telemetry should show hardware-monitor.service under a systemd system-unit directory. Legitimate overlap is resolved through ownership, content and change-control evidence.

KQL · UNTESTED HUNTING

```
DeviceFileEvents
| where FileName =~ "hardware-monitor.service"
| where FolderPath startswith "/etc/systemd/system/"
    or FolderPath startswith "/usr/lib/systemd/system/"
    or FolderPath startswith "/lib/systemd/system/"
| project Timestamp, DeviceId, DeviceName, ActionType, FileName, FolderPath,
    SHA1, FileSize, InitiatingProcessFileName,
    InitiatingProcessCommandLine, InitiatingProcessAccountName, ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	KEY	STATUS
None	Retention	None	Device + time + ReportId	Untested

Why Defender XDR

Direct file-event telemetry, documented path/process fields, strong DeviceId pivot and Live Response support. The query does not conflate Sentinel schemas.



TRIAGE & INVESTIGATION

Turn a filename match into a defensible disposition.

WORKFLOW

1 Confirm event

Verify device, exact path, timestamp, ActionType and writer process.

2 Retrieve unit

Use approved administration or Live Response; preserve hash, owner, mode and timestamps.

3 Validate ownership

Run `rpm -qf` or `dpkg -S`; compare CMDB and change records.

4 Inspect systemd

`systemctl cat/status/is-enabled hardware-monitor.service.`

5 Pivot

Review referenced executable/module, process tree, network events and adjacent files.

Escalate

Unknown owner, early-boot ordering, suspicious module or unexplained writer.

Benign

Versioned package, approved deployment, stable hash and documented service owner.

Inconclusive

File deleted, incomplete telemetry or no content retrieval. Preserve as exposure signal.

Containment boundary

Do not disable a production service on filename alone. Validate host role and ownership first unless corroborating compromise evidence requires emergency response.



FALSE POSITIVES & VALIDATION

What must be proven before alerting.

QUALITY CONTROL

SCENARIO	LIKELIHOOD	DISPOSITION
Local monitoring service with generic name	M	Validate owner/package/hash
Configuration-management deployment	M	Map writer and change ticket
Malicious SPECTRE persistence	L	Escalate with content/module pivots
Pre-existing file before onboarding	M	File event may be absent

Validation steps

1. Create a uniquely named benign test unit in each approved unit directory.
2. Confirm DeviceFileEvents and exact field population.
3. Run the campaign hunt across full retention.
4. Retrieve and disposition every production match.

Decision criteria

Success: controlled writes are visible and real matches are investigable.

Rejection: benign overlap dominates without robust ownership scoping.

Failure: controlled events do not arrive or path fields are absent.

Allowlisting rule

Allowlist only a versioned, owned package/deployment with stable path, hash or signer/package provenance and an approved change record. Never suppress by filename alone.



CONCLUSION & REFERENCES

Coverage statement, platform limits and document control.

FINAL

CONCLUSION

Useful exposure hunt, not a production detection.

The exact unit filename is a defensible, source-observed pivot. Its value is immediate prevalence assessment; its limitations are artifact dependence, unknown benign overlap and lack of file-content visibility.

Cross-SIEM decision

Splunk: no reliable implementation without verified Linux raw source, index, sourcetype and field extraction.

Chronicle: no reliable YARA-L rule without verified parser, UDM event type and file/process mappings.

Deferred ideas

acpi_pad.ko filename-only alert; encrypted HTTP path detection; named-pipe hunt; generic BYOVD duplication. Each lacks adequate specificity or verified schema for this dossier.

References

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Document control

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